RADBox_lite Equipment Startup Guide

To get the equipment up and running, it is necessary to do the following:

Step 0: Assemble RADBox_lite and configure the Arduino to start its operation

1. If this is the first time using the RADBox_lite equipment, it will be necessary to assemble and configure it beforehand. To do so, go to the repository using the following link: https://github.com/Sofializarazo/RADbox/tree/Radbox_proyect
2. Inside the "instructions" folder, you will find a file called "Setup Instructions", which contains detailed instructions for the assembly and initial configuration of the equipment.

If you already have the equipment assembled and the Arduino configured with the code, then proceed to step 1.

Step 1: Download of the Base Software

The program required to visualize the data produced by the equipment is hosted in a GitHub repository. Follow these instructions to obtain it:

1. Access the repository using the following link: https://github.com/Sofializarazo/RADbox/tree/Radbox_proyect
2. Go to the folder named app and download the file "RADBox_lite.exe" if you are using Windows, or "RADBox_litemac.command" if you are using a MacBook.
3. Once downloaded, save the file in a folder on your desktop for easy access.
   - If, when downloading the file on Windows, a security message appears indicating that the file comes from an unknown source or asking whether you want to continue with the download, select "Yes" or "Continue anyway". The file is safe. This warning may appear on some computers because the system does not recognize the origin of the file and, for security reasons, warns the user before opening it. In that case, simply confirm the download and continue with the installation process.
   - If, when trying to open the application on MacBook, a security message appears indicating that the file comes from an unknown source and cannot be opened securely because it cannot be verified for malware, do not worry. This is an additional security measure implemented by macOS. To resolve it, go to the Apple menu in the top-left corner, open System Settings, access Privacy & Security, and scroll down to the Security section on the right panel. There, you will see a message similar to: "RADBox_litemac.command was blocked because it is not from an identified developer." In that case, click

"Open Anyway", and the application will run correctly and macOS will recognize it as safe in future launches.

- Tip: If the system continues to block the application even after following the steps above, copy the application file (.exe or .command, as applicable) to a USB drive. Then transfer the file from the USB drive to your desktop instead of downloading it directly from the repository. In many cases, this prevents the operating system from marking the file as potentially unsafe, allowing it to run without additional permissions and making it automatically recognized as a safe file.

4. Do not change the source file name, as this may cause errors.

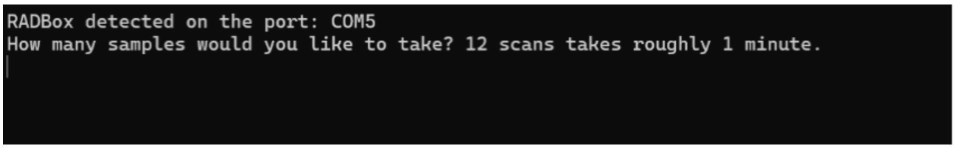
5. If you need to modify the software, refer to Appendix 1.

Step 2: Connection and System Execution

Once the drivers have been installed and the base software has been downloaded, the system is ready to operate:

1. Connect the RADBox_lite device to an available USB port on your computer.
2. Go to the folder where the RADBox_lite file was saved (RADBox_lite.exe or RADBox_litemac.command) and run the main program by double-clicking the executable file.
3. After opening the file, a command window or configuration interface will appear (it may take approximately one to one and a half minutes to appear):

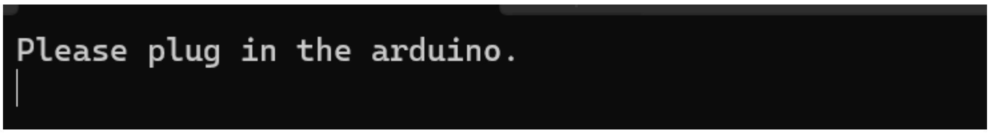
- For Windows, the data will be automatically saved in the folder where the application is located. Simply enter the number of data points you wish to collect and press Enter.



- For macOS, a more extensive warning message may appear. The system may indicate that Matplotlib is building the front cache; this may take a moment; in that case, ignore the message, as the application is starting correctly.
  Once the application loads properly, you will be asked to select the folder where you want to save the data. Choose the location that is most convenient for you.

Afterwards, the system will ask how many samples you wish to take. Enter the corresponding number and press Enter. At that point, data acquisition will begin automatically.

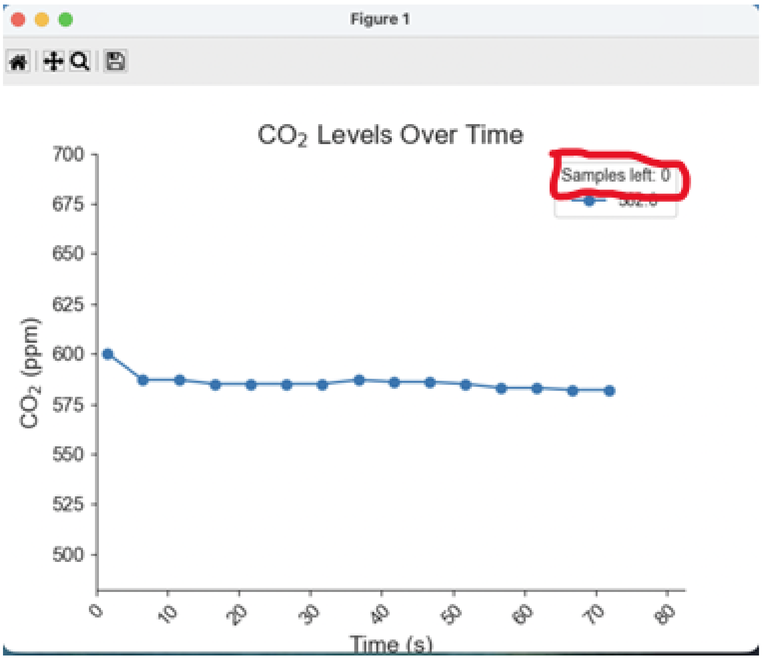
If it is disconnected, the following message will appear, and the program will proceed to close:



Disconnect it and reconnect it or plug it into a different USB port. If this does not work and the message continues to appear, refer to Appendix 2.

4. The system will automatically start data acquisition. Do not close the application until it has finished. The data will be plotted, showing trends and variations over time.

5. Once the sampling process is completed, the program will display: Samples left: 0.



6. You can safely close the application; there is no need to manually save it beforehand.

Step 3: Storage of Results

1. The data is automatically stored in the same folder where the executable file is located for Windows, or in the folder selected during data acquisition for MacBook. It is recommended to rename the generated file immediately after its creation to facilitate identification and future analysis.
2. The data is automatically exported in .csv format (comma-separated values). To view it after disconnecting RADBox_lite, simply double-click the generated file to open it with Microsoft Excel.
3. In most cases, this is not necessary. However, if the data does not appear in column format when opening the file, follow these steps:
   - Go to the following link: https://convertio.co/es/csv-xlsx/
   - Select your file and click the button to convert it to .xlsx format.
   - Once the process is complete, download the file. When opened in Excel, the data will be correctly displayed in organized columns.
4. When opening the file, the data will be arranged in columns, ready to be processed or analyzed.

	A	B
1	Time (s)	CO2 (ppm)
2	0.83	638
3	5.86	650
4	10.9	662
5	15.93	660
6	20.98	661
7	26.04	640
8	31.09	633
9	36.13	631
10	41.16	631
11	46.21	631
12	51.26	628
13	56.31	628
14	61.36	627
15	66.4	626
16	71.45	621

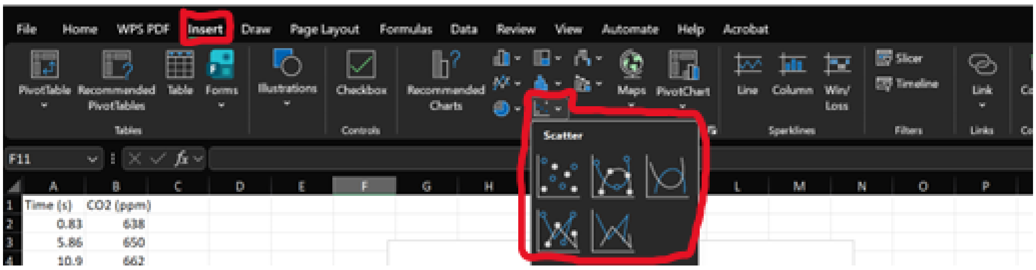
5. To recreate the visual trends in Excel, follow the procedure described below:
   - Select the columns of data that you want to plot.

	A	B	C
1	Time (s)	CO2 (ppm)	
2	1.56	600	
3	6.57	587	
4	11.59	587	
5	16.61	585	
6	21.63	585	
7	26.64	585	
8	31.66	585	
9	36.68	587	
10	41.7	586	
11	46.71	586	
12	51.72	585	
13	56.74	583	
14	61.76	583	
15	66.78	582	
16	71.79	582	
17			

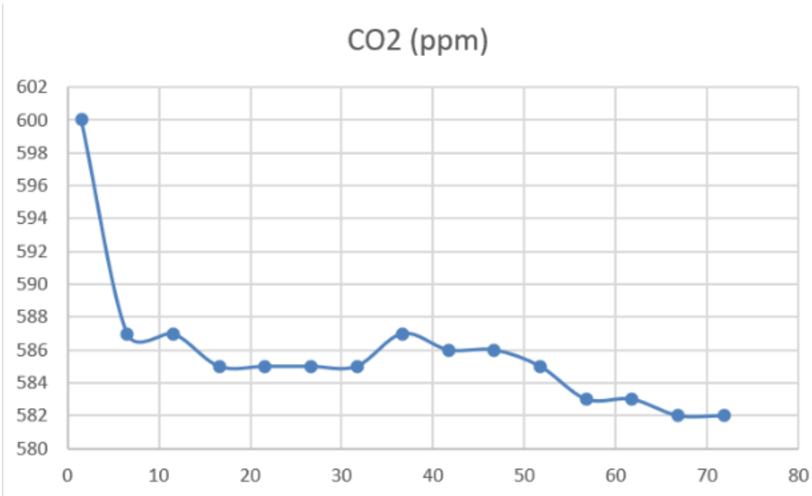
}

   - Go to the Insert tab in the top ribbon.
   - Click on the Scatter icon  . This will generate a graph where each point represents an individual reading, allowing you to observe the distribution of the data.

- If you prefer to visualize the progression using lines to better identify the behavior of the variables, select the chart options with trend lines 



- The data plotted in Excel will look as follows:



Appendix 1: Guide for Editing the Python Code

This annex is intended for users who wish to explore the internal functioning of the software, add custom features, or run the program directly from the source code.

The file with the .py extension (source code) is located in the same GitHub repository, inside the source_code folder. Go there and download the file RADBox_lite.py for Windows and RADBox_litemac.py for MacBook. To properly view and edit it, follow these instructions:

On windows:

1. Development Environment Setup

Although multiple text editors exist, for this manual we will use Visual Studio Code (VS Code) for its versatility and integrated tools.

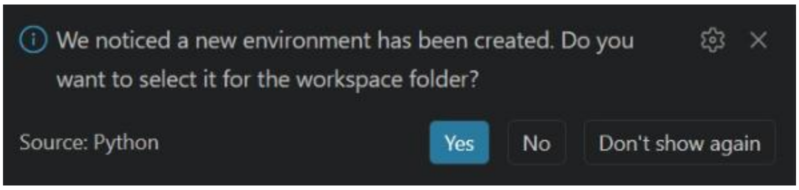
1. Download and install the editor from the official site: code.visualstudio.com .
2. To ensure code compatibility, Python version 3.11.9 is required; therefore:
    a. Go to: python.org/downloads/windows.
    b. Locate and download the installer: “Download Windows installer (64-bit)”

    c. During installation, make sure to check the box “Add Python to PATH” before completing the process; if this option is not shown, simply accept all permissions.

2. Visual Studio Code Preparation

Once both programs are installed, configure the editor as follows:

1. Copy your .py file and paste it into a new folder created exclusively for these project modifications.
2. Open Visual Studio Code, go to the top menu, and select File > Open Folder.
3. Search for and select the folder you just created, then click Select Folder.
4. In the left panel (Explorer), click on your file to open it in the editor.
5. To interact with the language, you must enable the integrated terminal; to do this, go to the top menu and select View > Terminal.
6. In the terminal that appears at the bottom, you can confirm that the system recognizes the python installation correctly; to do this, type the following command and press Enter: “python –version” You should see a message confirming the version: Python 3.11.9.
7. To avoid conflicts with other projects, create an isolated environment with the specific version of Python by typing: “py -3.11 -m venv Radbox”.
8. If a notification appears in the bottom right corner asking if you want to use the new environment for the folder, select Yes.



9. Execute the following command to start working within the project environment: ".\Radbox\Scripts\activate"
10. Confirm that the environment is active when the text (Radbox) appears in green at the beginning of the command line.

3. Library Installation

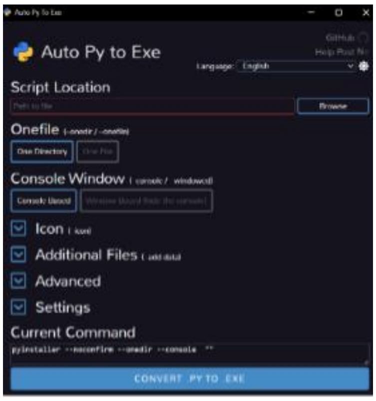
For the code to function correctly, it is necessary to install the libraries that manage communication with the equipment and the generation of graphs.

1. Copy and paste the following command to install the necessary libraries: "py -m pip install pyserial matplotlib seaborn"
2. To verify that the process was successful, type: "py -m pip list". Ensure that pyserial (v3.5 or higher), matplotlib (v3.6.3 or higher), and seaborn (v0.12.1 or higher) appear in the list.
3. The code is ready to be edited to your liking.

4. Converting Code (.py) to Executable (.exe)

Once you have made the desired edits to the code, you can convert it into a standalone application to run it without the need to open the code editor.

1. In the terminal write "py -m pip install auto-py-to-exe" and press Enter.
2. Once the download is finished, type the following command in the same window: "py -m auto_py_to_exe"
3. A graphical interface will open.



Configure the fields as follows:

a. Script Location: Click on “Browse” and select your file named RADBox_lite.py or the name you have given it.
b. Onefile: Select the “One File” option to generate a single executable file.
c. Console Window: Select “Console Based”.
d. Settings: In “Output Directory”, define a new folder where you want to save your final application; your data will also be saved in this same folder.

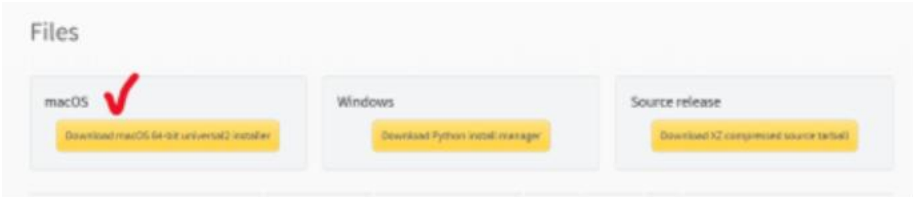
4. Click the blue button Convert .py to .exe. Once the process is finished, your new executable file will be ready to operate with the RADBox_lite equipment.

On MacBook:

1. Development Environment Setup

Although multiple text editors exist, for this manual we will use Visual Studio Code (VS Code) for its versatility and integrated tools.

1. Download and install the editor from the oficial site: https://code.visualstudio.com/Download. Download for Mac
2. To ensure code compatibility, Python version 3.11.9 is required; therefore:
    a. Go to: https://www.python.org/downloads/release/python-3119/.
    b. Locate and download the installer: “Download macOS 64-bit universal2 installer”



c. During installation, make sure to check the box “Add Python to PATH” before completing the process; if this option is not shown, simply accept all permissions.

2. Visual Studio Code Preparation

Once both programs are installed, configure the editor as follows:

1. Copy your .py file and paste it into a new folder created exclusively for these project modifications.
2. Open Visual Studio Code, go to the top menu, and select File > Open Folder.
3. Search for and select the folder you just created, then click Select Folder.
4. In the left panel (Explorer), click on your file to open it in the editor.
5. To interact with the language, you must enable the integrated terminal; to do this, go to the top menu and select View- terminal.
6. In the terminal that appears at the bottom you can confirm that the system recognizes the Python installation correctly; to do this, type the following command and press Enter: “python3.11 --version”. You should see a message confirming the version: Python 3.11.9.
7. To avoid conflicts with other projects, create an isolated environment with the specific version of Python by typing: “python3.11 -m venv Radbox”.
8. Execute the following command to start working within the project environment: “source Radbox/bin/activate”
9. Confirm that the environment is active when the text (Radbox) appears at the beginning of the command line.

3. Library Installation

For the code to function correctly, it is necessary to install the libraries that manage communication with the equipment and the generation of graphs.

1. Copy and paste the following command to install the necessary libraries: “pip install pyserial matplotlib seaborn”
2. The code is ready to be edited to your liking.
3. To run the code, type “python3 RADBoxmac.py”

4. Converting Code (.py) to Executable

Once you have made the desired edits to the code, you can convert it into a standalone application to run it without the need to open the code editor.

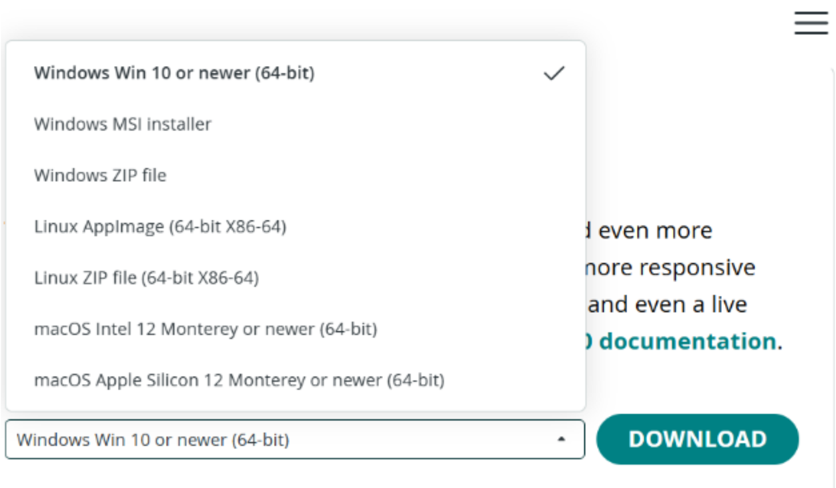
1. In the terminal, type the following:
   "pip install pyinstaller"
2. Now enter:
   "pyinstaller --noconfirm --onefile --console --name="RADBox_litemac.command" RADBox_litemac.py"
3. Then: "dist/RADBox_litemac.command && chmod +x dist/RADBox_litemac.command"
4. And that's it; the executable file will be saved in the folder inside another folder called dist.

Appendix 2: Arduino Installation

Some computers do not have the necessary dependencies for the system to correctly recognize an Arduino device. In these cases, the RADBox_lite application may fail to detect that the device is connected and will display the message "plug in Arduino", even when the device is physically connected.

The solution to this issue is simple: the Arduino environment must be properly installed on the computer. To do so, follow the steps below:

1. Go to the following link: https://www.arduino.cc/en/software
2. Select the version compatible with your computer and click on download.



3. Wait for the installation to finish and open the application. Accept all additional installations or permissions that are requested, as these are necessary for the system to correctly recognize the Arduino when it is connected to the computer.
4. Then, connect the device and reopen the Arduino application. Verify that no additional installations are requested, which will confirm that the system is now correctly recognizing the device.
5. To check this, with the device connected, go to Tools > Ports within the Arduino application and verify that one of the available ports appears as Arduino Nano Every.
6. Finally, close the Arduino application and open RADBox_lite (.command or .exe depending on the operating system used). At this point, the equipment will be ready to use.